

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 20
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 4
forecast_steps: 48
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-08-01 15:56:51

Training Start: 2025-08-01_14-44-46

Training End: 2025-08-01_15-28-43

Training Duration: 43.95 minutes

Predictions Start: 2025-08-01_15-28-43

Predictions End: 2025-08-01_15-42-46

Predictions Duration: 14.05 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.2741

MSE: 2.4990

RMSE: 1.5808

R2 Score: 0.8617

Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0149

MSE: 0.0003

RMSE: 0.0183

R2 Score: 0.9174

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 85,269 rows

First 5 timestamps BEFORE:

1744397085760000
1744397085770000
1744397085789000
1744397085801000
1744397085817000

...

Last 5 timestamps BEFORE:

1744464971366000
1744464971377000
1744464971390000
1744464971402000
1744464971414000

First 5 timestamps AFTER:

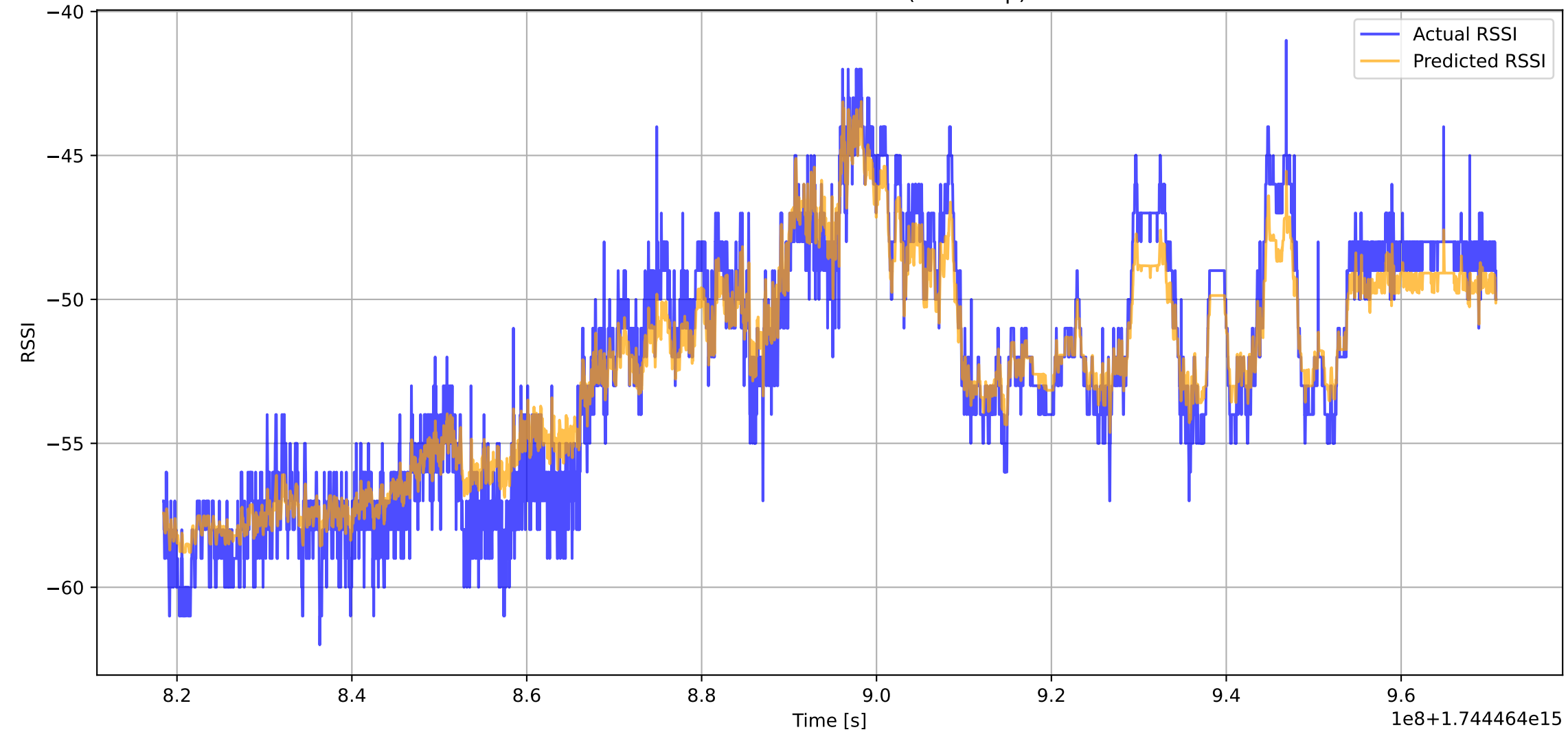
1744397085760000
1744397085770000
1744397085789000
1744397085801000
1744397085817000

...

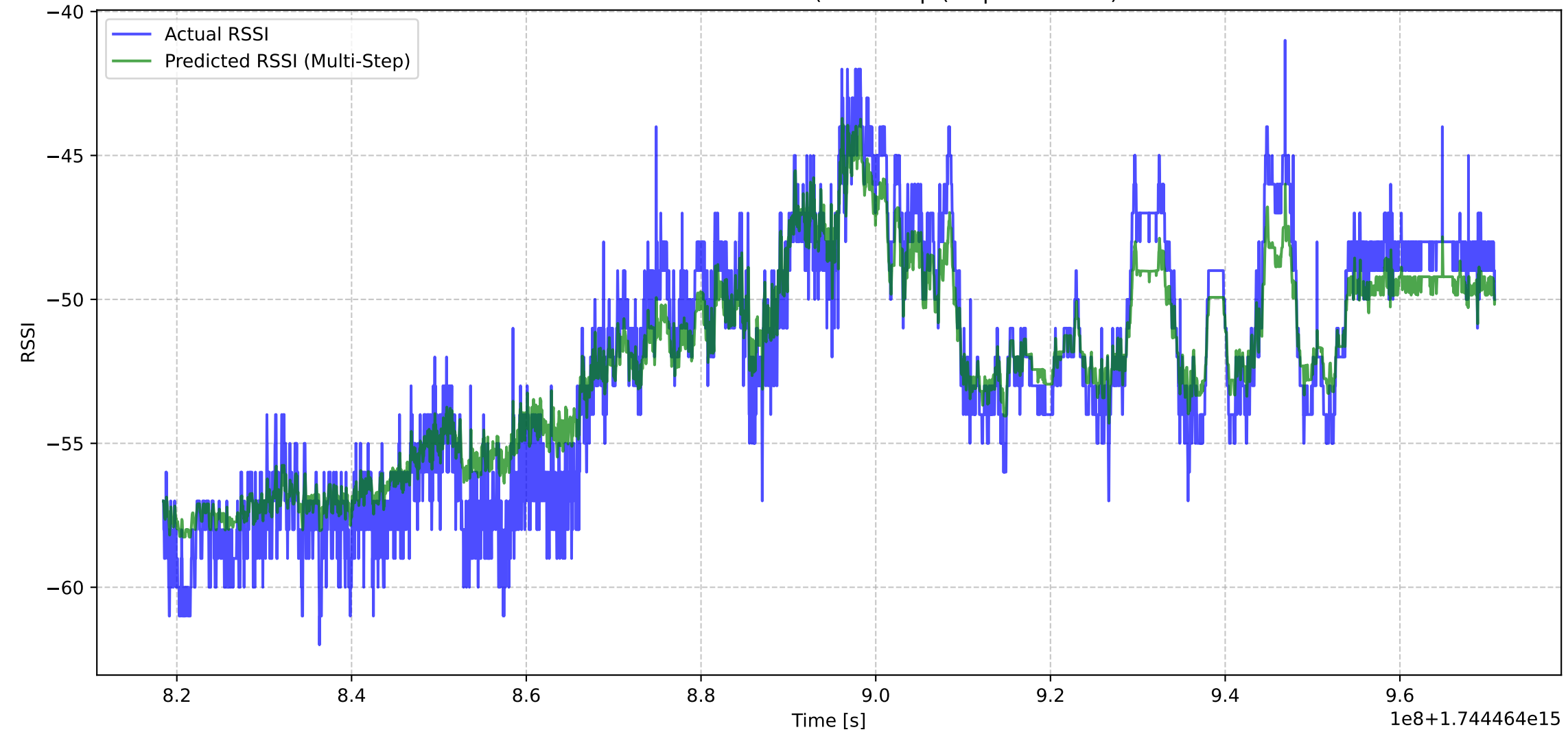
Last 5 timestamps AFTER:

1744464971366000
1744464971377000
1744464971390000
1744464971402000
1744464971414000

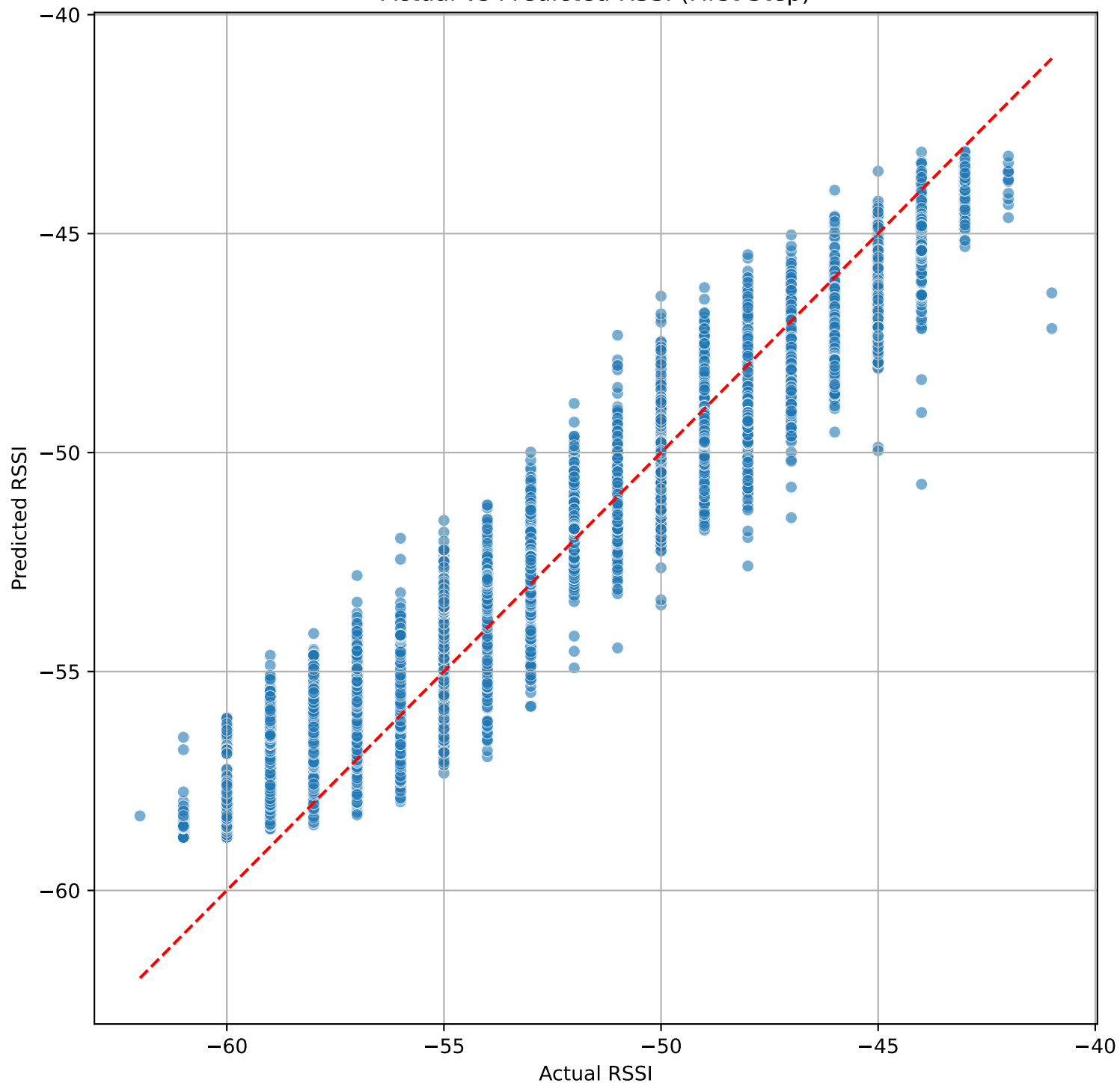
Actual vs Predicted RSSI (First Step)



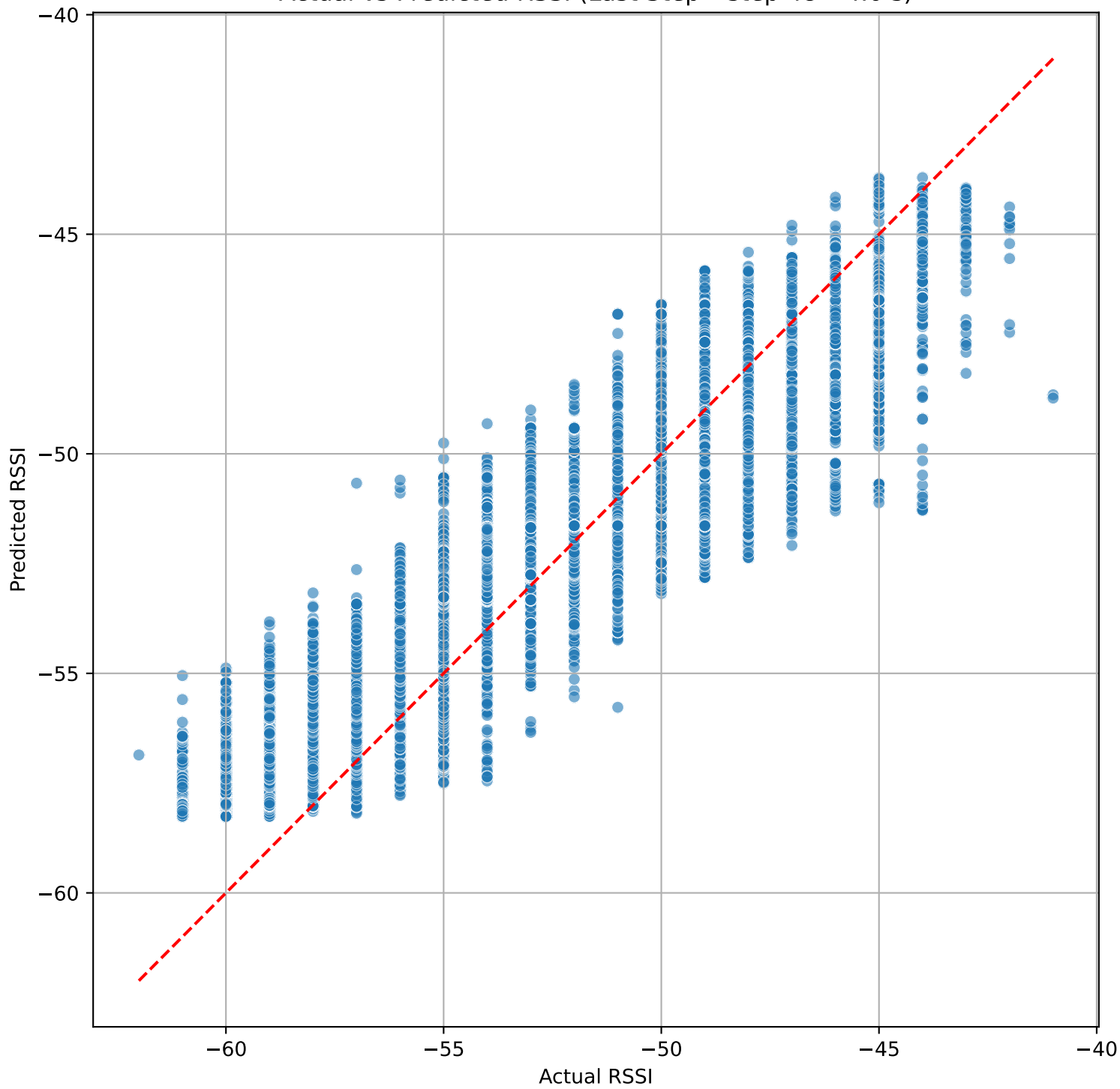
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$ 4.0 s))



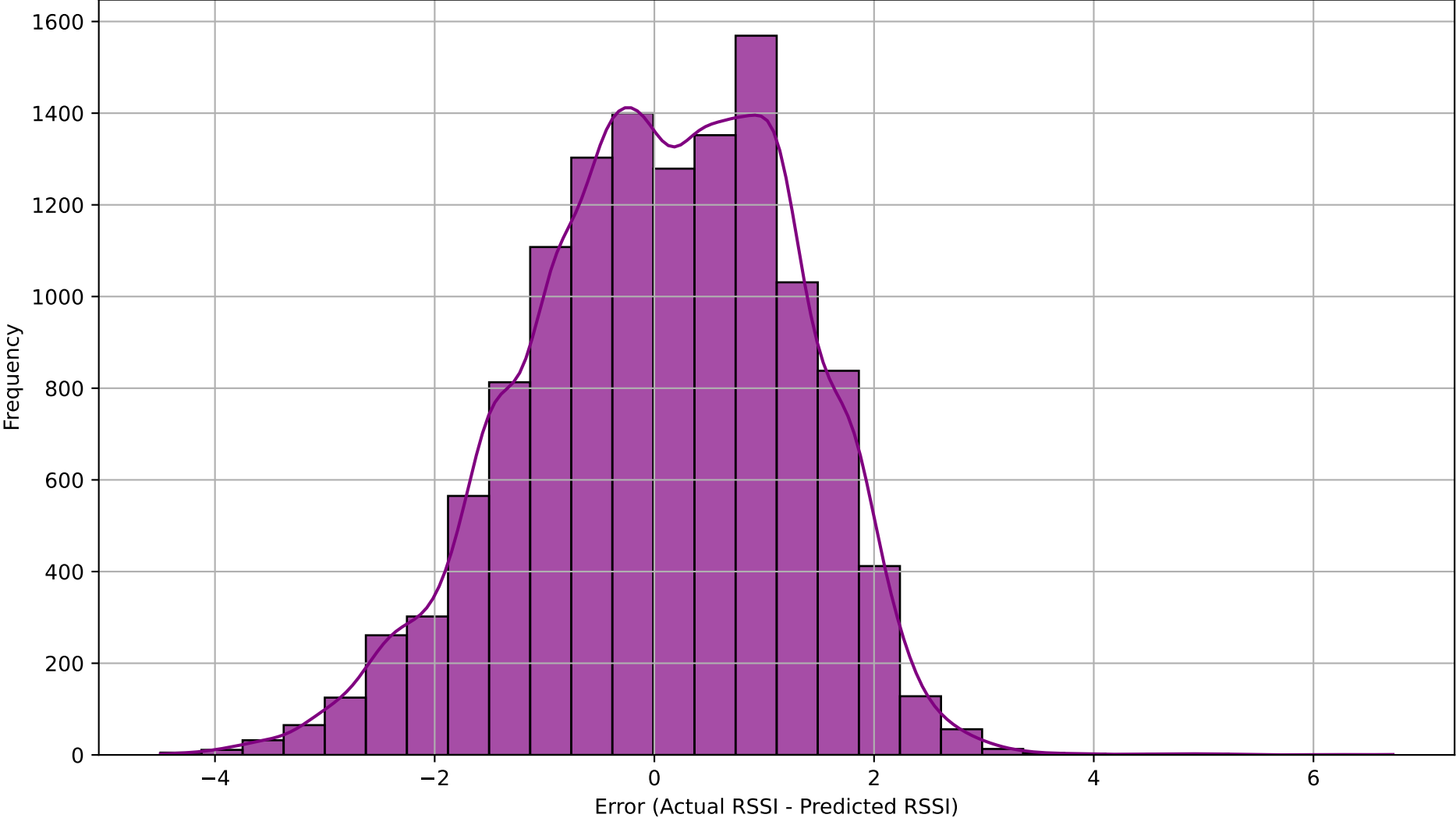
Actual vs Predicted RSSI (First Step)



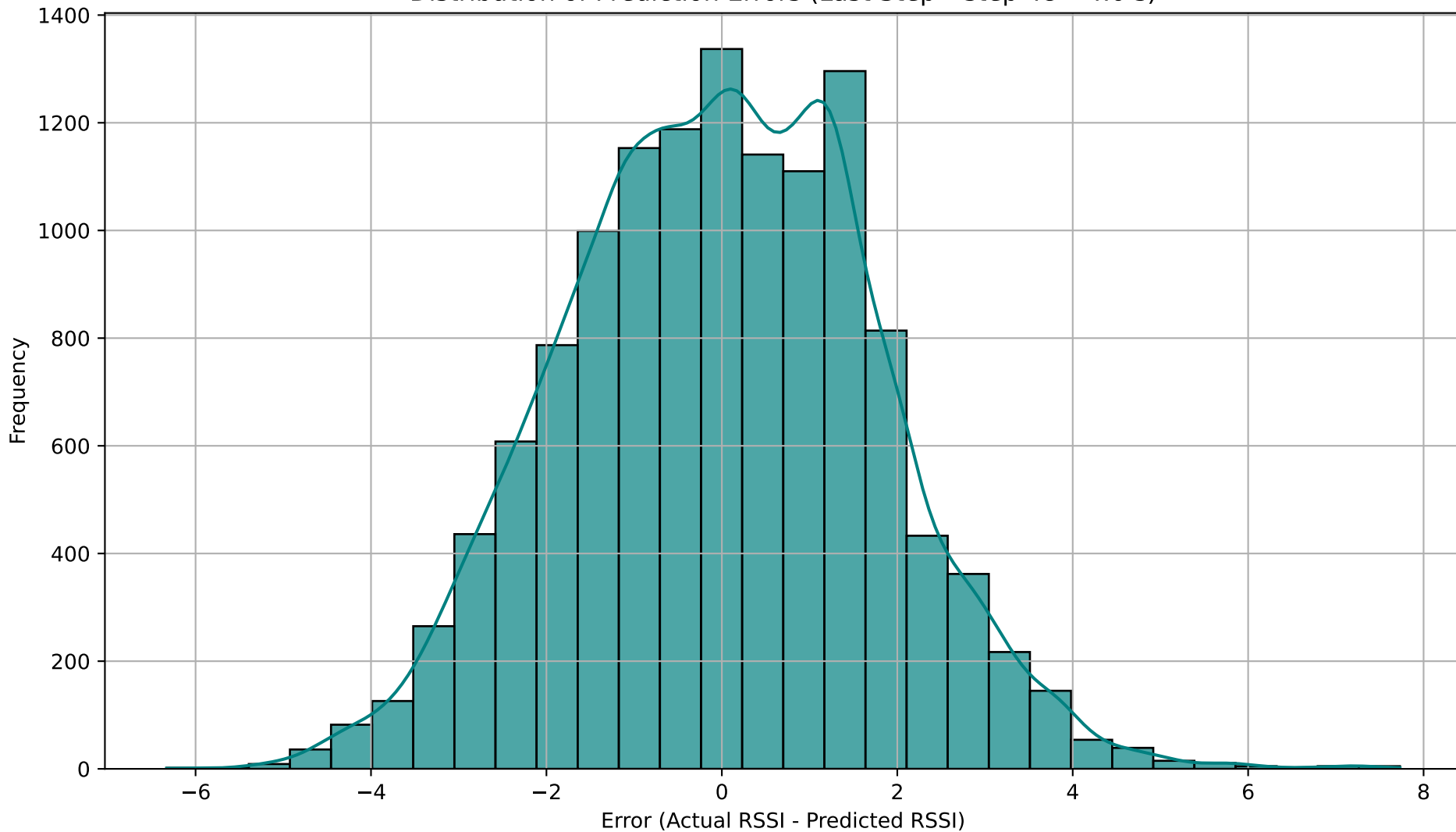
Actual vs Predicted RSSI (Last Step - step 48 $\approx$ 4.0 s)



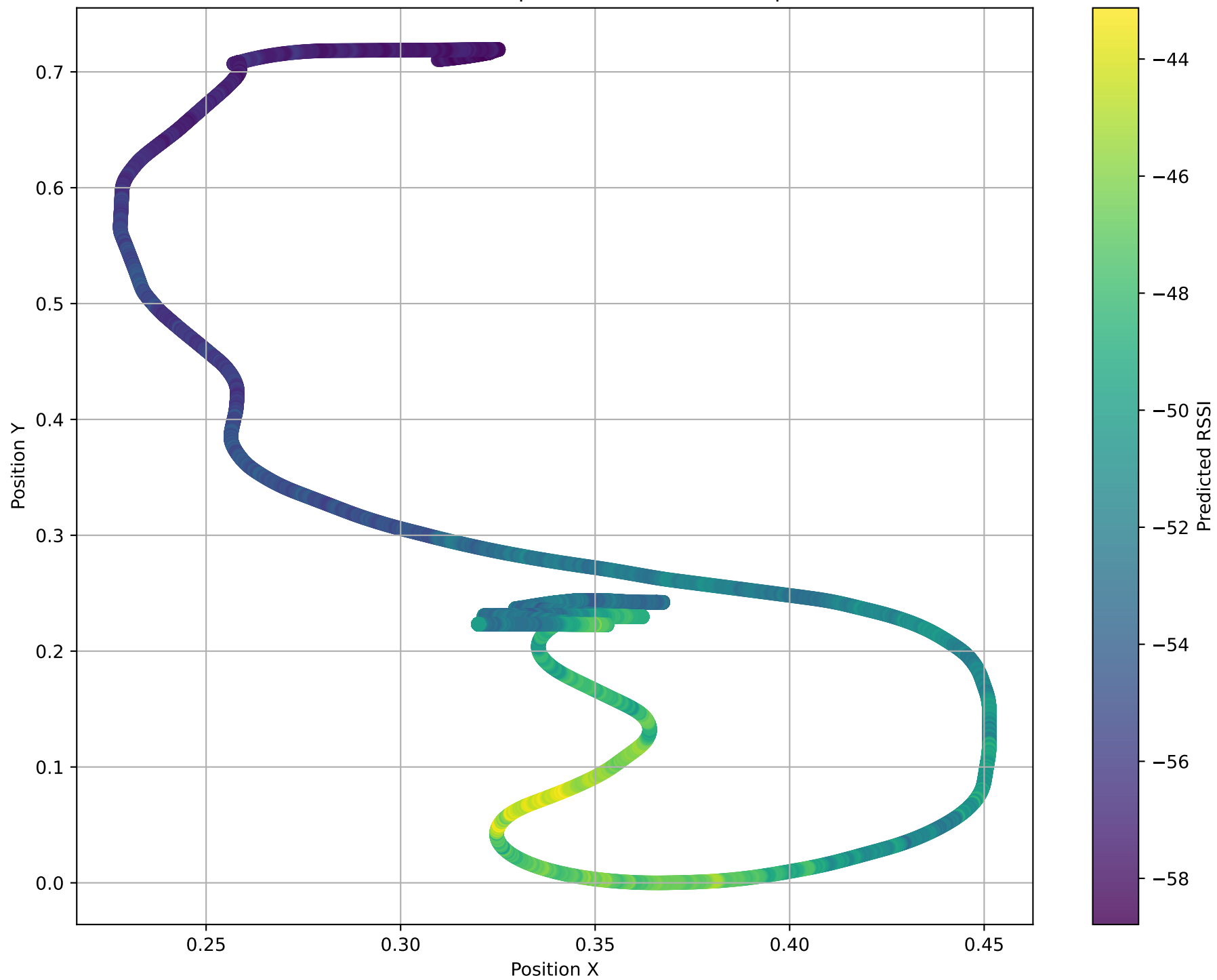
Distribution of Prediction Errors (First Step)



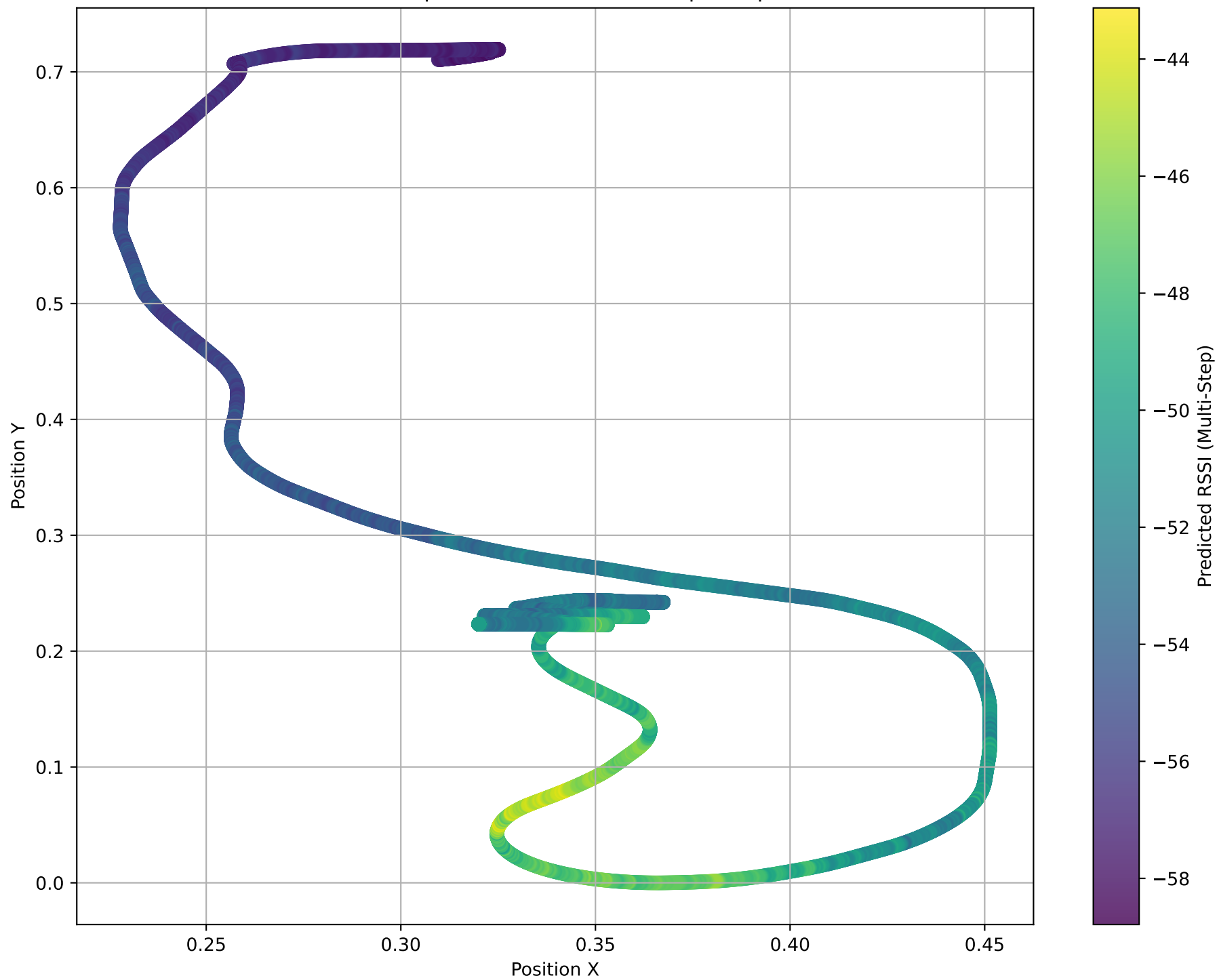
Distribution of Prediction Errors (Last Step - step 48 $\approx$ 4.0 s)



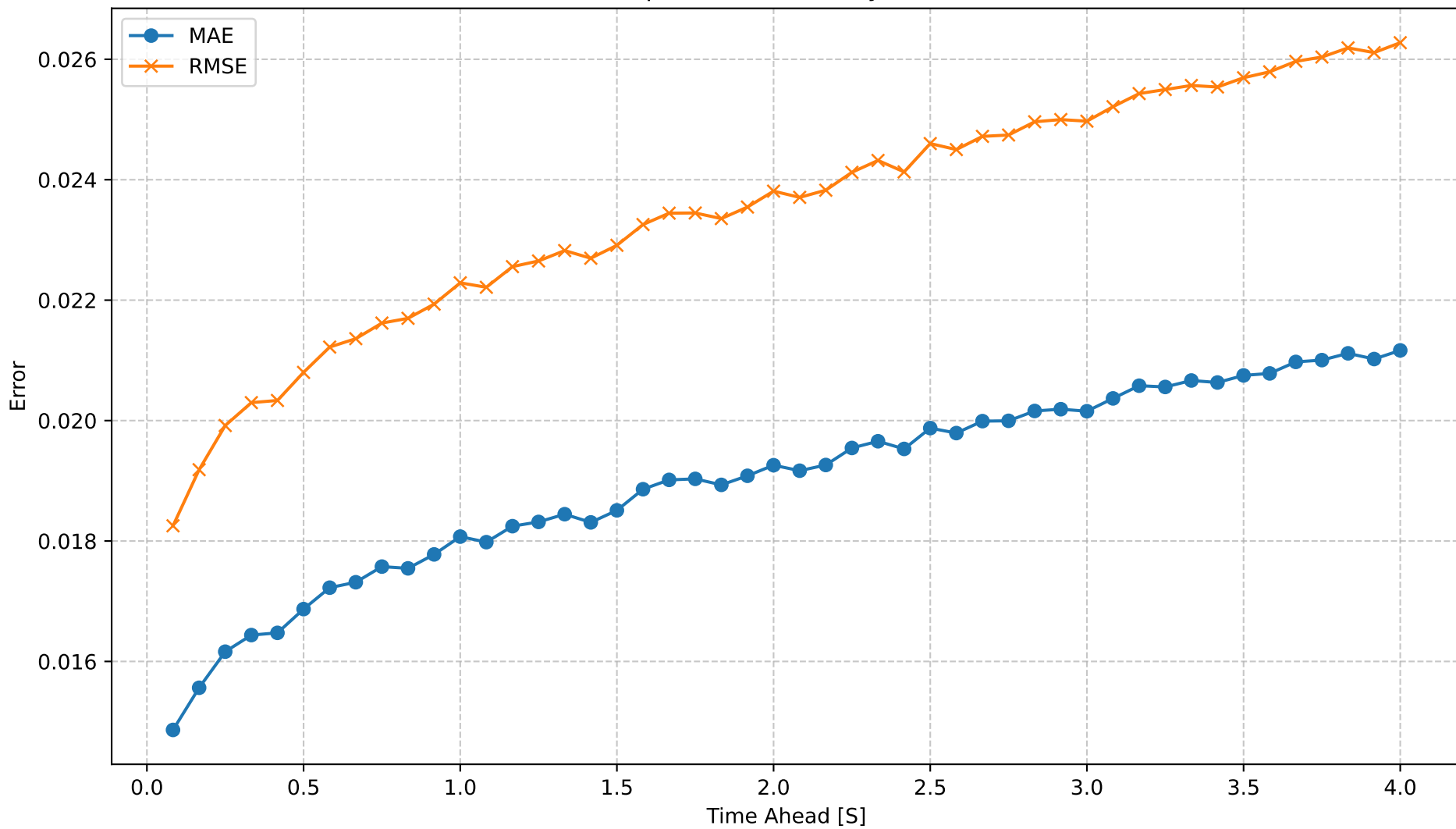
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.02
- Mediana : 0.02 / 0.02
- 48º passo : 0.02 / 0.03

Crescimento do erro: 42.4% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 48 (0.02), 46 (0.02), 47 (0.02)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.